Master EIT ICT Labs in ICT Innovation

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Non textual information retrieval

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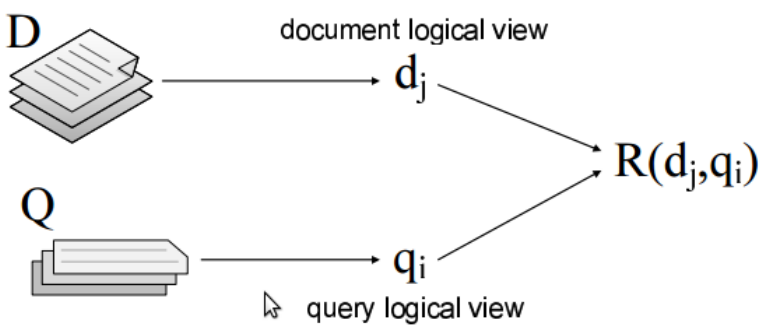
Goals of visual retrieval

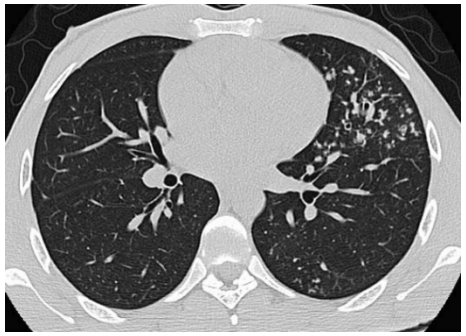
- Retrieval of images by visual (objective) means
  - Features extracted fully automatically
    - Colours,
    - Textures,
    - Shapes,
    - Interest points, …
- Query by image example, QBE
  - “find me images visually similar to this one”

IR Models

- An **IR model** is a quadruple $[\mathbf{D}, \mathbf{Q}, \mathcal{F}, R(q_i, d_j)]$ where
  1. $\mathbf{D}$ is a set of logical views for the documents in the collection
  2. $\mathbf{Q}$ is a set of logical views for the user queries
  3. $\mathcal{F}$ is a framework for modeling documents and queries
  4. $R(q_i, d_j)$ is a ranking function



CBIR Architecture

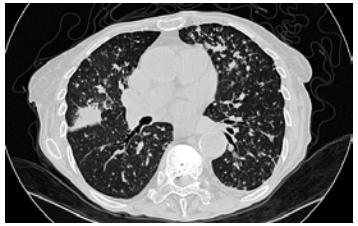


Represented by

Colour 1
Colour 2
…
Texture N
…

Stored in

User queries



Colour 1
Colour 2
…
Texture N
…

Database

Feedback

CBIR: Content based image retrieval

- CBIR require define how we can index and retrieve images based on its contents
- While textual documents consists of terms, sentences and, in some extensions: hyperlinks; images require a bit more complex processing
- There are mainly four different ways of searching images, metadata

CBIR: Content based image retrieval

- The first one is searching for images based on attributes
  - such as file name,
  - date of creation
  - manually added categories, such as subject and creator.

  - This can be made using relational model and searches are made based on these attributes

- This method don’t care about image content

CBIR architecture

CBIR generalities

- Each CBIR system may have a different underlying query model, but the basic issues are the same:
- Image Feature Information: Color, Texture and Shape, etc
- High-Dimensional Indexing Structures: R-tree, SS-tree, etc
- Similarity Measures: Euclidean Distance, Weighted Euclidean Distance, etc
- Query Processing Algorithm: K-nearest neighbor search, Range Search, etc

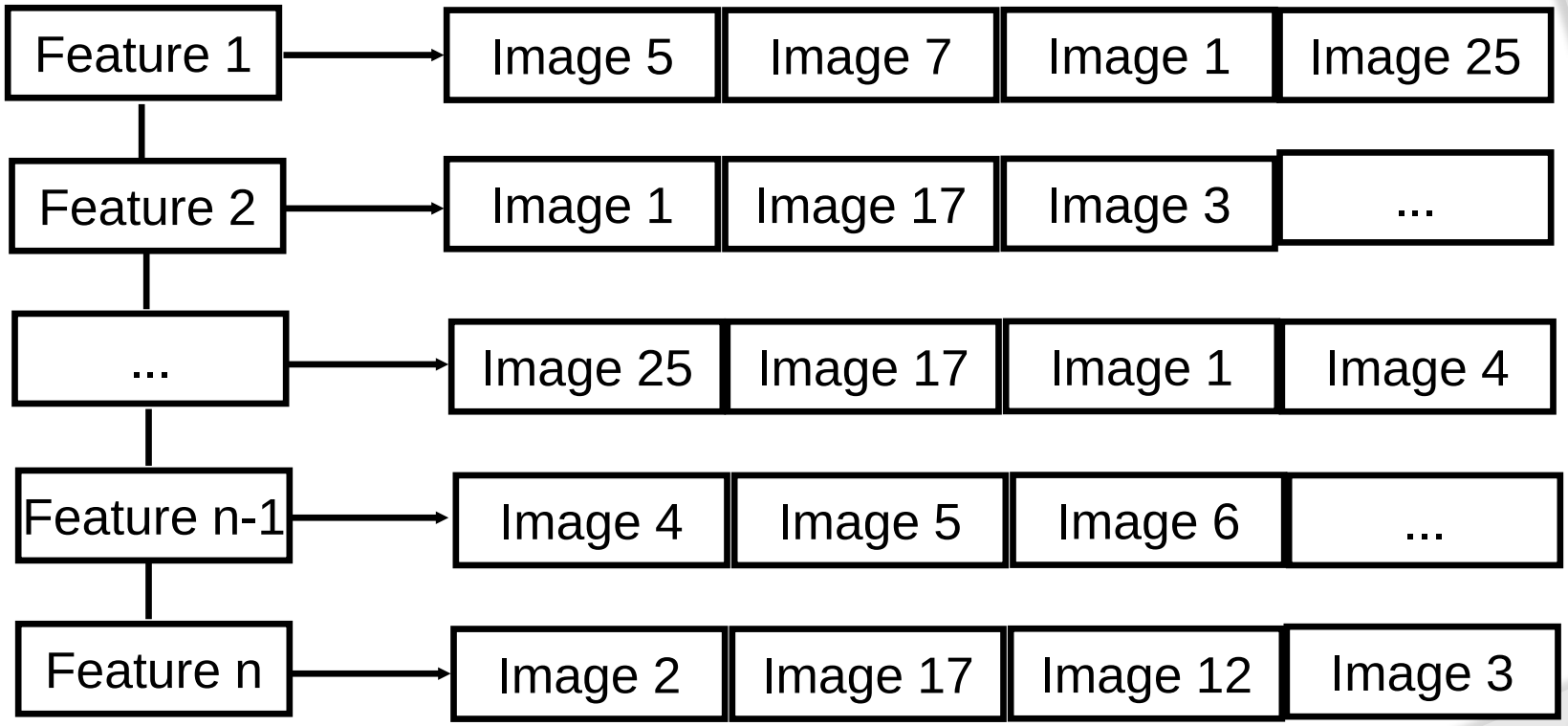
General view on the concept of feature

- Features are numerical values computed from each image (continuous values)
  - View connected to image classification
  - Ideas and methods from classification and machine learning
  - Inspired by k-nearest neighbor approach

General view on the concept of feature

- One to One matching
  - Different distance functions depending on comparing feature
  - Euclidean Distance, Weighted Euclidean Distance
- One to Many matching
  - Features are image properties that are present or absent
    - View connected to textual information retrieval
    - Ideas and methods from text retrieval
- ~85’000 possible features, 1’000-3’000 features per image, distribution similar to words in text

The inverted file



Some CBIR Systems

- In this section, we provide brief details of the following ‘historical’ CBIR systems
  - QBIC (IBM)
  - Virage (UC San Diego/Virage)
  - Blobworld (UC Berkeley)
  - VisualSEEk (Columbia)
  - Photobook (MIT)
  - MARS (Urbana, UC Irvine)
  - LIRE
  - Google Images
  - TinEye

QBIC

Usage: I: Get Info C: Histogram L: Layout T: Find Similar Texture S: Special Hybrid Color

Query was:
Random

Google reverse search: WSABIE

- Most popular search engines such as Yahoo and Google use an altered mix of image descriptors, image objects and basic image data

- They use things as file names, size and dates, but also analyses the HTML page the image is in. The
  - Google bots cannot read the image, so an alternative text is provided instead.
  - Using image ALT tag, image Filename, surrounding text, overall theme

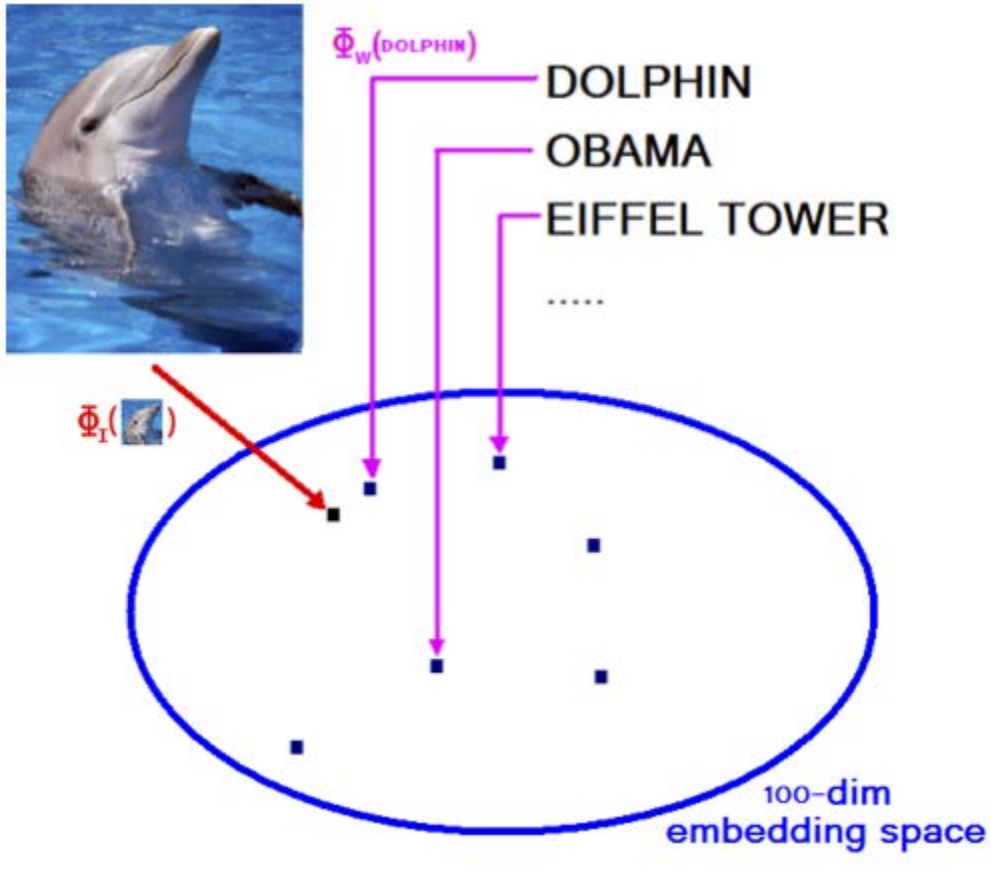
Google reverse search: WSABIE

- Annotate images using textual descriptions found in HTML

- Associate image descriptors:
  - There is no official available description of algorithms
  - Typical image fingerprinting for image matching:
    - SIFT is slow and not good at illumination changes, while it is invariant to rotation, scale changes and affine transformations.
    - SURF is fast and has good performance as the same as SIFT, but it is not stable to rotation and illumination changes.
    - PCA-SIFT is the best but it has problems with image blur.

Google reverse search: WSABIE

- Joint Word-Image Model



Google reverse search: WSABIE

- Ability for:
  - textual to image search
  - Image to image search
  - Image to text search

- https://www.google.es/imghp?sbi=1

Google reverse search: WSABIE

- WSABIE: Scaling Up To Large Vocabulary Image Annotation
  Proceedings of the Twenty-Second International Joint Conference on Artificial Intelligence
  Barcelona
  July 19, 2011